

Project of Discrete Event Systems - A.Y. 2024/25

This document describes the steps to follow for the project of Discrete Event Systems. The project can be done individually or in a group, but forming groups of three/four members is highly recommended.

1. Find a real discrete event system that you would like to model. Though it is not mandatory, it would be nice to motivate your choice with a picture, a video, a scheme, a datasheet, etc.
2. Contact the instructor for a meeting where you will be asked to illustrate your idea. The instructor will help you define the “boundaries” and some simplifying assumptions for the proposed system so that the resulting discrete-event model is of reasonable size to be managed for a homework project.
3. Prepare the state automaton model for the system. When ready, send it to the instructor for a final check about the coherence of the assumptions and the size. In this phase the instructor will not evaluate the correctness of the model (this will be done during the discussion).
4. Address the following points.
 - (a) Assume a Poisson clock structure.
 - i. Compute the limit state probabilities analytically. If possible, the rates of the exponential distributions should be chosen so that all the limit state probabilities are approximately of the same order of magnitude.
 - ii. Estimate the limit state probabilities using simulations over a time horizon such that the steady state is reached and compare them with the true values. Choose the number of simulations so that the magnitude of the estimation error is less than 10^{-3} for all state probabilities.
 - iii. Give evidence of the Law of Large Numbers showing, by means of tables and/or figures, the sample mean and the sample variance of the estimated probabilities for different orders of magnitude of the number of samples used to compute a single estimate.
 - (b) Assume a “realistic” clock structure for the considered system.
 - i. Verify whether the system admits steady state by estimating the state probabilities versus time and showing whether they converge to limit values.
 - ii. In case of a positive answer to the previous question, repeat point 4(a)iii.
5. Prepare a short report where you describe the system and the simplifying assumptions made, the state automaton model, the theoretical tools used, the simulation/estimation techniques applied, and the results obtained. Please use figures and/or tables to explain the results.
6. Send the report and the Matlab code to the instructor to schedule the date for the discussion. In case of a group project, all the members of the group must attend the discussion. Notice that, during the discussion, each member of the group could be asked to describe portions of the Matlab code, and the procedures implemented therein.